



UNIVERSITY OF THE PUNJAB

Roll No.

Fifth Semester 2017
Examination: B.S. 4 Years Programme

PAPER: Sampling Techniques (Theory)
Course Code: STAT-305

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE

Q. No. 1: Encircle the correct option.

1. When the sample survey becomes a census survey, the sampling error becomes -----
A. Zero B. One
C. Constant D. None of the above
2. For a population that is not normally distributed, the distribution of the sample means will -----
A. be negatively skewed B. approach the normal distribution.
C. be positively skewed. D. never approaches normal distribution.
3. The process of drawing a sample from a population is known as -----
A. Census B. Survey
C. Sampling D. None of above
4. Suppose we select every fifth invoice in a file. What type of sampling is this?
A. Simple Random B. Cluster
C. Stratified D. Systematic
5. Which of the following is not an example of random sampling method:
A. Simple random Sampling B. Stratified Sampling
C. Cluster Sampling D. Convenience Sampling
6. All possible samples of size 'n' are selected from a population and the mean of each sample is determined. What is the mean of the sample means?
A. Exactly the same as the population mean B. Larger than the population mean
C. Smaller than the population mean D. Cannot be estimated in advance
7. Sampling in which sampling unit can be repeated more than once is called -----
A. Sampling with replacement B. Sampling without replacement
C. Both (A) & (B) D. None of the above
8. ----- is the standard deviation of a sampling distribution.
A. Sample standard deviation B. Replication error
C. Meta error D. Standard error
9. A complete list of all the units in the population is called -----
A. Sampling unit B. Population unit
C. Sampling frame D. None of the above
10. If a sample is drawn from each stratum minimizing the probabilities of non-preferred samples, it is known as:
A. Selection with proportional allocation B. Controlled selection
C. Haphazard selection D. None of the above



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TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE

Attempt all questions.

- Q. No. 2. Write a short note on the following: (5 each)
- Inverse Sampling
 - Method of controlled selection in Stratified Random Sampling
 - Reasons of Stratification
 - Application of Systematic Sampling

- Q. No. 3. If the loss function due to an error in $\bar{y}$ is $d(\bar{y} - \bar{Y})^2$ and the cost function is $C = C_0 + C_1 n$, then show that the most economical value of 'n' in simple random sampling, ignoring finite population correction is $\sqrt{\frac{dS^2}{C_1}}$. (07)

- Q. No. 4. Prove that sample proportion is an unbiased estimator of population proportion with variance (07)

$$V_{pst} = \frac{1}{N^2} \sum \frac{N_h^2(N_h - n_h)}{N_h - 1} \frac{P_h Q_h}{n_h}$$

- Q. No. 5. If the terms in $\frac{1}{N_h}$ are not ignored then prove that (09)

$$V_{ran} = V_{prop} + \frac{1-f}{n(N-1)} \left\{ \sum_{h=1}^L N_h (\bar{y}_h - \bar{y})^2 - \frac{1}{N} \sum_{h=1}^L (N - N_h) S_h^2 \right\}$$

- Q. No. 6. Show that the mean of a systematic sample is more precise than the mean of a simple random sample if and only if $S_{wsy}^2 > S^2$ (07)
where S_{wsy}^2 and S^2 are variance within systematic samples and population variance respectively.